

Learning to Rescale: On-the-Fly Sequence Length Adaptation in Non-Autoregressive Speech Synthesis

Jiawei Jin^{1,*}, Ren Wang^{1,*}, Zhiyu Cui², Shun Lei¹, Yixuan Zhou¹, Zhiyong Wu^{1,**}

¹ Shenzhen International Graduate School, Tsinghua University, China

² Ant Group, China

jjw23@mails.tsinghua.edu.cn, zywu@sz.tsinghua.edu.cn

Abstract

Non-autoregressive (NAR) text-to-speech (TTS) models excel in parallel inference and style consistency. However, their reliance on accurate character-level or global duration predictions remains a critical bottleneck, limiting both synthesis fidelity and naturalness. This paper proposes ElasticDLM (Elastic-length Diffusion Language Model), a novel variable-length NAR TTS model that eliminates the strong dependency on accurate duration predictions. We introduce two functional tokens, along with a Differentiated Length-Scaling Training Scheme (DLTS) and a Hierarchical Confidence-Guided Inference (HCGI) strategy tailored for speech characteristics, enabling the model to learn length-scaling features and automatically adjust sequence length during inference. Experiments show that ElasticDLM produces high-fidelity, natural speech from arbitrary-length inputs, demonstrating a more flexible and simplified approach to high-quality speech synthesis. Audio samples are available at <https://thuhcsi.github.io/ElasticDLM/>.

Index Terms: Speech Synthesis, Non-Autoregressive Models, Variable-Length Synthesis

1. Introduction

Text-to-Speech (TTS) synthesis [1, 2, 3] aims to generate high-fidelity, natural speech for diverse applications like intelligent assistants [4], audiobooks [5], and assistive communication systems [6]. The core challenge for TTS lies in learning the precise alignment between text and complex acoustics. By decomposing the complicated global mapping task into simple, local conditional predictions, autoregressive (AR) models [7, 8] effectively tackle the alignment challenge via their sequential, token-wise generation process. However, their token-by-token nature inherently leads to slow inference and exposure bias [9, 10, 11].

To address these AR limitations, non-autoregressive (NAR) models [12, 13, 14] have emerged as a compelling alternative, prized for their parallel inference and superior global style consistency. However, the parallel generation paradigm of NAR models confronts the challenge of sequence alignment. Early strategies relied on character-level durations, typically derived from methods like teacher-forcing with alignments distilled from AR models [15, 16], monotonic alignment search (MAS) [17, 18] or targets from the Montreal Forced Aligner (MFA) [19]. These pre-extracted durations are often time-consuming and inaccurate, especially under noisy or in-the-wild speech conditions, severely compromising the training. Moreover, this reliance on a rigid and ‘fine-grained’ alignment hinders the models from achieving human-like naturalness and rich stylistic diversity.

To overcome the shortages of character-level duration modeling, emerging works such as E2-TTS [20], F5-TTS [21], and MaskGCT [22] have adopted masked generative modeling to adaptively distribute word or phoneme duration according to a total sequence length. Nonetheless, these fixed-length methods still rely on predicting the accurate global duration as an initial sequence length. This mechanism fundamentally restricts the flexibility and accuracy, leading to two major failures. Firstly, inaccurate duration predictions result in audible artifacts such as unnatural speaking rates and fragmented prosody. Secondly, this rigid duration paradigm prevents the model from adaptively adjusting the generated sequence length with different semantic information and acoustic characteristics. These flaws hinder the practical application [23] of the NAR-based TTS models.

Recently, several studies in text generation [24], most notably DreamOn [25], have attempted to address the issue of variable-length sequence generation based on NAR models. Diffusion Language Models (DLMs) [26] are widely adopted, mainly leveraging its single-token-level insertion and deletion capability. However, it is non-trivial and challenging to directly apply this paradigm to the speech synthesis domain. Firstly, due to the high temporal resolution of speech, where a single phoneme often corresponds to multiple semantic tokens, and the inherent acoustic redundancy, discrete single-token operations like merging or insertion fail to provide the discernible and robust gradient for learning complicated length manipulations. Secondly, the vast length of speech sequences clashes with the limited inference budget, making it a formidable challenge to reconcile the need for large-scale length variability with the demand for high-fidelity generation. To tackle all these problems and overcome the fixed-length constraint in existing NAR speech synthesis frameworks, we propose ElasticDLM: a novel variable-length NAR speech synthesis approach. The main contributions of this work are as follows:

- (1) We propose a novel variable-length NAR speech synthesis model. To eliminate reliance on accurate duration predictions, it introduces two functional tokens for unified, learnable length regulation, enhancing robustness and flexibility.
- (2) We develop a Differentiated Length-Scaling Training Scheme (DLTS) tailored to speech characteristics, which combines block-wise merging and insertion operations with an auxiliary length-scaling loss, providing the model with salient and learnable features for length regulation within information-sparse acoustic sequences.
- (3) We propose a Hierarchical Confidence-Guided Inference (HCGI) strategy adapted to the high dynamic range of speech sequences, enabling simultaneous length refinement and content generation within a coarse-to-fine process to ensure both length flexibility and high fidelity.

*These authors contributed equally.

**indicates the corresponding author.

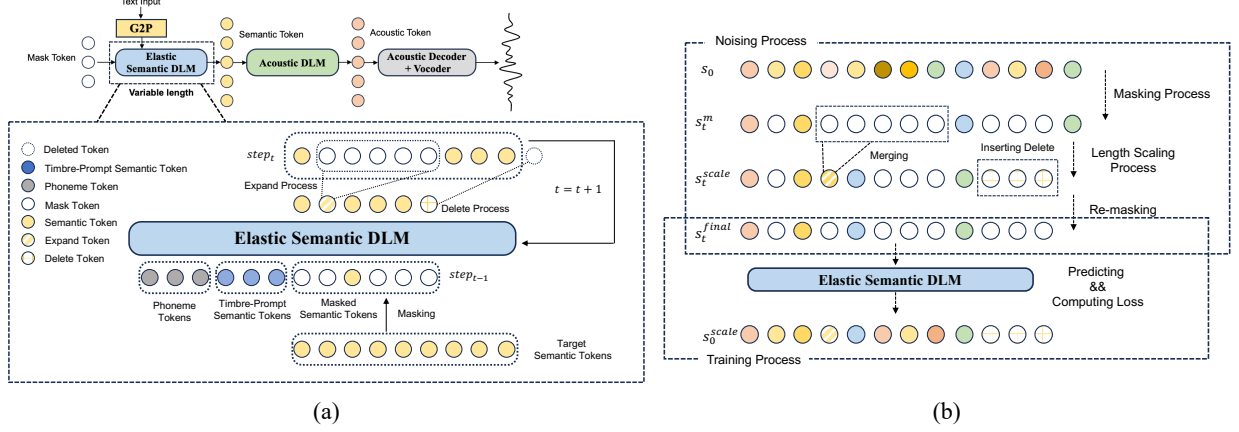


Figure 1: Overall architecture of the proposed method (a) and design details of the Differentiated Length-Scaling Training Scheme (b).

2. Proposed Method

2.1. System Overview

Our proposed non-autoregressive TTS framework (Fig.1(a)) employs a cascaded Diffusion Language Model (DLM) to progressively refine text into speech via iterative mask prediction. The system is driven by our novel Elastic Semantic DLM, which first maps text to semantic tokens. These tokens are then converted into speech by a subsequent acoustic modeling stage (Acoustic DLM, Acoustic Decoder, and Vocoder) based on the MaskGCT [22] architecture.

As depicted in the lower part of Fig.1(a), our Elastic Semantic DLM iteratively refines an initial mask sequence (derived from phonemes and a timbre prompt) into a semantic token sequence. Uniquely, it incorporates [EXPAND] and [DELETE] functional tokens, which respectively trigger the insertion of new mask tokens or the removal of existing segments, allowing for dynamic length adjustment. This capability is supported by two key components: a Differentiated Length-Scaling Training Scheme (DLTS, Fig.1(b)) to enable the model to learn salient length-scaling features, and a Hierarchical Confidence-Guided Inference strategy (HCGI, Algorithm 1) to ensure both high length flexibility and synthesis fidelity during generation.

2.2. Differentiated Length-Scaling Training Scheme

As shown in Fig.1(b), the DLTS defines a speech-tailored noising process operating on an initial token sequence s_0 at a given diffusion step t . The process begins by applying standard content masking, yielding a partially masked sequence $s_t^m = \text{Mask}(s_0, t)$. This is followed by differentiated length scaling, which employs two inverse noising operations to create distinct training targets. To overcome the inherent acoustic redundancy and extract explicit contraction signals from information-sparse speech sequences, we replace simple pairwise merging with a stochastic compression of larger blocks. Specifically, each non-overlapping block of n consecutive masked tokens is independently merged into a single representation with probability p , formally defined as:

$$s_t^{\text{scale}} = \text{Merge}(s_t^m, n, p) \quad (1)$$

Conversely, to teach the model to utilize [DELETE] tokens for sequence reduction, the process explicitly appends k [DELETE] tokens to the end of the sequence, defined as:

$$s_t^{\text{scale}} = \text{InsertDelete}(s_t^m, k) \quad (2)$$

Here, the number of inserted tokens k is sampled from $k \sim \text{Uniform}(1, \lfloor \alpha(t) \cdot L \rfloor)$, where L is the sequence length and $\alpha(t)$ is a time-dependent scaling factor. These two inverse processes are applied either separately or simultaneously during training and the differentiated construction prevents functional confusion between [EXPAND] and [DELETE] tokens. Furthermore, by coupling the operations with the diffusion timestep t , our scheme aligns with the hierarchical nature of speech, encouraging major structural length adjustment early and fine-grained acoustic refinements later, enhancing both the robustness and interpretability. Finally, all [EXPAND] and [DELETE] tokens are re-masked to [MASK] state as $s_t^{\text{final}} = \text{Re-mask}(s_t^{\text{scale}})$.

The model is trained to predict the clean scaled sequence s_0^{scale} , given the final noised sequence s_t^{final} and conditional information C including phoneme tokens and the timbre-prompt semantic tokens, which is modeled as the conditional probability $p_\theta(s_0^{\text{scale}} | s_t^{\text{final}}, C)$. The model's parameters θ are optimized by minimizing a composite loss function. Because addressing speech sparsity necessitates these block-wise operations, the cost of mispredicting functional tokens is heavily amplified. Therefore, we augment the standard negative log-likelihood (NLL) loss. This term enforces consistency between the predicted length change and the ground-truth change. The total loss $\mathcal{L}$ is formally defined as:

$$\mathcal{L} = \mathbb{E}_{s_0, t} \left[- \sum_{i \in \mathcal{M}_t} \log p_\theta \left((s_0^{\text{scale}})_i | s_t^{\text{final}}, C \right) + \lambda \sum_{i \in \mathcal{M}_t} |(\Delta \hat{L}_i) - \Delta L| \right] \quad (3)$$

where the first term is the NLL loss over all masked positions $\mathcal{M}_t$. The second term is the length loss, weighted by a hyperparameter λ , which minimizes the absolute error between the ground-truth length change ΔL and the total expected length change predicted by the model. The expected change at each position i is $\Delta \hat{L}_i = (n-1) \cdot p_\theta([\text{EXPAND}]_i) - p_\theta([\text{DELETE}]_i)$. This composite objective forces the model to learn not only the correct token identities but also the overall structural length modifications.

2.3. Hierarchical Confidence-Guided Inference Strategy

The sinusoidal scheduler in conventional DLM inference brings a fundamental challenge for variable-length speech synthesis. It stifles essential length adjustment in the early stages by predicting too few tokens, while encouraging high-risk, large-scale corrections later when the iteration budget is nearly de-

Algorithm 1: One-Step Inference in HCGI

Input: $s_{t-1}^m, m_{t-1}, t, L_{\text{curr}}, \tau$, Model M, Expand number n

Output: $s_t^m, m_t, L_{\text{new}}$

$L \leftarrow M(s_{t-1}^m, m_{t-1}, t)$;

$p \leftarrow \text{Softmax}(L)$;

for $j \leftarrow 0$ **to** $\text{length}(p - 1)$ **do**

if $p_{\text{exp}}[j] > \tau$ **then**

$\text{Ops}[j] \leftarrow \text{EXPAND}$

else if $p_{\text{del}}[j] > \tau$ **then**

$\text{Ops}[j] \leftarrow \text{DELETE}$

Mask out tokens: $L_{\text{content}} = L[:, \text{Exp \& Del}] \leftarrow -\infty$;

$s_t \leftarrow \text{Top-k-Sample}(L_{\text{content}})$;

$s_t^m \leftarrow \text{Re-mask}(s_t, N_{\text{mask}})$;

for j **in** Ops **do**

if $\text{Ops}[j] = \text{DELETE}$ **then**

 Delete $s_t^m[j]$

else if $\text{Ops}[j] = \text{EXPAND}$ **then**

 Replace $s_t^m[j]$ with $n \times [\text{MASK}]$

$L_{\text{new}}, m_t \leftarrow \text{UpdateLengthAndMask}(s_t^m)$;

return $s_t^m, m_t, L_{\text{new}}$;

pleted. Such belated, disruptive changes corrupt delicate acoustic structures and introduce audible artifacts, making it impossible to achieve both accurate duration and high-fidelity speech. To address this acoustic mismatch, we introduce the speech-tailored HCGI strategy to effectively balance dynamic length scaling with high-fidelity generation. Crucially, as a sampling-stage modification, HCGI introduces negligible latency overhead compared to the backbone’s forward pass.

As shown in Algorithm 1, within each given denoising step t , our method performs a unified prediction to yield the logit distribution $L = M(s_{t-1}^m, m_{t-1}, t)$, which is conditioned on the previous sequence s_{t-1}^m and mask m_{t-1} . Next, the confidence scores p_{exp} for [EXPAND] and p_{del} for [DELETE] tokens are extracted from the softmax of L , any functional token whose confidence score exceeds a threshold τ is immediately sampled and retained for execution:

$$s_t^{\text{scale}} = \text{Ops} = \{j \mid p_{\text{exp}}[j] > \tau \text{ or } p_{\text{del}}[j] > \tau\} \quad (4)$$

This step facilitates the rapid sampling of high-confidence functional tokens, enabling effective length adjustment early in the denoising process. Simultaneously, top-k sampling is performed using the “content” logits L_{content} , which are created by masking out sampled tokens. Following this, the N_{mask} tokens with the lowest confidence are re-masked, where N_{mask} is determined by the scheduling function $N_{\text{mask}} = \lfloor \gamma(t+1) \cdot L_{\text{curr}} \rfloor$. Here, γ is a sine scheduling function that determines the masking ratio and L_{curr} represents the current number of effective tokens. Finally (Fig.1(a)), each [EXPAND] is replaced by n [MASK] tokens, and [DELETE]s are removed. Given this altered dimension, `UpdateLengthAndMask` explicitly recalculates the new length L_{new} and mask m_t for step $t+1$. This iterative refinement dynamically adapts length while progressively solidifying semantic content into high-fidelity output.

3. Experiments

3.1. Experiment Setup

Our Elastic Semantic DLM is built upon the 16-layer Transformer architecture (hidden size 1536, 16 attention heads) of MaskGCT’s Text-to-Semantic model and is initialized by loading the pre-trained weights of its Transformer component. The model is trained on 100k hours of data from the English and Chinese subsets of Emilia-large. The training incorporates our

proposed noising strategy, where [EXPAND] and [DELETE] operations are applied independently with a 0.5 probability, with the [EXPAND] operation specifically merging blocks of size $n = 5$. The entire training is conducted on 24 NVIDIA A100 (80GB) GPUs, optimized with AdamW using a $1e-4$ learning rate and 32K warmup steps. For inference, we use a default of 50 steps and a confidence threshold τ of 0.8. Sampling is performed using top-k ($k = 20$), where the temperature linearly anneals from 1.5 to 0 throughout the process.

3.2. Experiment Results

3.2.1. Main Results

We conducted our evaluations on the seed-test-zh and seed-test-en benchmarks from Seed-TTS, using MaskGCT as our primary baseline. To isolate and highlight the benefits of eliminating accurate duration, we designed three distinct settings for the MaskGCT baseline. The first, termed `Normal`, represents the standard MaskGCT which relies on its built-in duration model. The second, `Fix`, simulates a poor duration estimate by initializing the sequence to a fixed length (4s for Chinese, 4.5s for English). Finally, to serve as a performance upper bound, the GT setting provides MaskGCT with the ground-truth audio length.

Our objective metrics include Word Error Rate (WER) for accuracy, L1-distance on sequence length (Len-L1) for variable-length capability, and WavLM-based cosine similarity for speaker similarity (Sim). For subjective evaluation, we conducted a Mean Opinion Score (N-MOS) test for naturalness (prosody and speech rate), where 15 professional listeners rated 10 random samples per test set. All results are presented with 95% confidence intervals.

As shown in Table 1, our proposed model (Ours-Normal) exhibits strong performance. It achieves WER comparable to the standard MaskGCT and even its oracle GT setting (slightly higher in Chinese, lower in English) with nearly identical Sim, indicating that our variable-length capability does not compromise fundamental synthesis quality. More importantly, our model secures the best results across both testsets on the Len-L1 and N-MOS metrics. This demonstrates that our model can effectively correct for potentially biased initial length predictions, yielding an output sequence length closer to the ground truth. This improved length control directly translates to more natural prosody and speech rate, as reflected by the superior N-MOS scores. In contrast, the original MaskGCT lacks this corrective mechanism; once an inaccurate length is predicted, it often leads to global or local unnaturalness in the final output.

Further evidence of our model’s superiority is found in the `Fix` scenario. Our model shows only a minor performance variation between the `Normal` and `Fix` settings across all key metrics (WER, Len-L1, N-MOS). Conversely, the baseline MaskGCT is highly sensitive to the initial length, showing a severe performance drop in the `Fix` case. This demonstrates that our model is independent on an accurate duration predictor and possesses a robust mechanism for dynamic length adjustment.

3.2.2. Ablation Study

We conducted ablation studies on the challenging `Fix` setting to validate our key designs (Table 1). First, removing the Length-Loss significantly increased duration error (Len-L1 from 0.8752s to 1.1129s on Chinese) and WER, confirming its crucial role in learning token semantics for prosody control. Second, disabling the HCGI strategy caused a severe performance collapse, with WER soaring to 4.005% (Chinese) and

Table 1: Main results on Seed-Eval test sets. WER and Len-L1 are ‘lower is better’ ($\downarrow$). Sim and MOS are ‘higher is better’ ($\uparrow$).

Model	seed-test-zh				seed-test-en			
	WER $\downarrow$	Len-L1 $\downarrow$	N-MOS $\uparrow$	Sim $\uparrow$	WER $\downarrow$	Len-L1 $\downarrow$	N-MOS $\uparrow$	Sim $\uparrow$
MaskGCT-GT	2.286%	-	4.23 $\pm$ 0.13	0.769	3.833%	-	4.27 $\pm$ 0.14	0.716
MaskGCT-Fix	4.182%	1.6993s	3.72 $\pm$ 0.16	0.771	13.341%	1.0156s	3.58 $\pm$ 0.19	0.718
MaskGCT-Normal	2.330%	0.8555s	3.84 $\pm$ 0.18	0.773	3.936%	0.5429s	3.96 $\pm$ 0.14	0.715
Ours-Fix	2.582%	0.8752s	4.17 $\pm$ 0.15	0.771	4.349%	0.6956s	4.29 $\pm$ 0.12	0.718
Ours-Normal	2.387%	0.7091s	4.20 $\pm$ 0.13	0.772	3.534%	0.5283s	4.44 $\pm$ 0.12	0.715
- w/o Length-Loss-Fix	2.798%	1.1129s	3.73 $\pm$ 0.15	0.763	4.530%	0.7231s	3.95 $\pm$ 0.12	0.714
- w/o HCGI-Fix	4.005%	1.5564s	3.47 $\pm$ 0.15	0.749	5.652%	0.7304s	3.05 $\pm$ 0.17	0.713

Table 2: WER comparisons under different initial length.

Model	1s	4s	7s	10s	13s	16s
E2-TTS [20]	95.91%	3.441%	2.873%	3.215%	6.236%	23.631%
F5-TTS [21]	73.82%	2.586%	1.804%	2.676%	5.991%	19.734%
MaskGCT	94.13%	4.182%	2.983%	16.245%	53.26%	107.764%
Ours	3.477%	2.582%	2.576%	2.447%	2.613%	2.459%

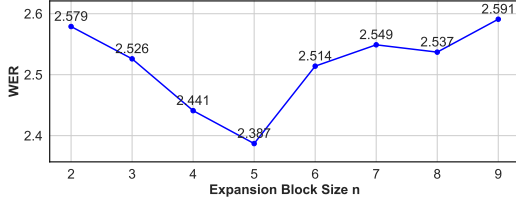


Figure 2: The WER with Different Expansion Block Size n

5.652% (English). These findings demonstrate that both components are indispensable for achieving robust performance, especially in a challenging predictor-free scenario.

3.2.3. Analysis of Different Initial Length

As shown in Table 2, experiments on the seed-test-zh benchmark with initial lengths ranging from 1s to 16s demonstrate the exceptional robustness of our proposed model. First, when compared to the primary baseline MaskGCT, our model maintains a consistently low WER across the entire spectrum, whereas MaskGCT demonstrates a heavy reliance on accurate duration prediction. Furthermore, we compare our approach with more powerful NAR models, specifically E2-TTS [20] and F5-TTS [21]. Although their continuous representation-based architectures and superior pre-trained foundations bring better overall performance and a certain degree of robustness, their synthesis quality still deteriorates rapidly as the input length varies. Ultimately, all baseline models fail completely at extreme lengths (1s, 16s). Experiments also conducted on English test set yield similar results. These findings confirm our model’s powerful dynamic length adjustment capability, proving reliable under challenging duration-prediction-free conditions.

3.2.4. Analysis of Expansion Block Size n

We performed an ablation study to determine the optimal Expansion Block Size (n), a critical and sensitive hyperparameter controlling the n -to-1 merging (noising) and 1-to- n expansion (denoising). As shown in Fig.2, the WER on seed-test-zh

exhibits a U-shaped curve as n varies, indicating degradation when n is extreme. A small n fails to provide a strong compression signal, causing the model to treat expansion as noise, while a large n results in rigid, inflexible expansions during inference. Consequently, we identified $n = 5$, a value approximating the average phoneme duration, as the optimal setting, as it best balances accuracy and length-adjustment flexibility to achieve the minimum WER, with similar result on English data.

3.2.5. Analysis of Application Flexibility

Highly expressive speech exhibits extreme prosodic variations, making variable-length modeling flexibility urgently necessary. To evaluate this, we fine-tuned our model and the MaskGCT baseline on a highly expressive single-speaker dataset. An AB preference test was then conducted by 15 professional listeners, who evaluated synthesized held-out texts across four emotional styles: Fearful, Sad, Coquettish, and Excited.

Results (Fig.3) reveal our model’s significant advantage in rendering rich emotions through dynamic speaking paces. Conversely, MaskGCT, constrained by its external duration predictor, produced rigid paces, yielding prosody that mismatched emotional intents and degraded synthesis quality. These findings validate that our variable-length architecture endows the model with superior expressiveness and flexibility.

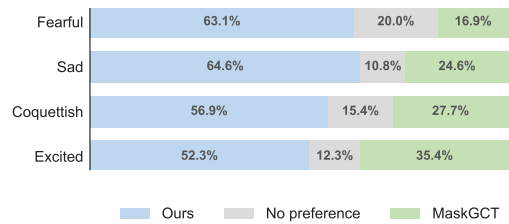


Figure 3: AB-Test on Highly Expressive Single-speaker Dataset

4. Conclusion

We proposed ElasticDLM, a novel variable-length NAR speech synthesis model driven by two functional tokens for unified length regulation. Tailored to inherent acoustic properties, it leverages DLTS to extract explicit length-scaling features from information-sparse speech sequences, and HCGI to ensure structural flexibility and acoustic fidelity through a coarse-to-fine inference process. Experiments validate that ElasticDLM effectively corrects erroneous initial duration estimates, delivering robust, high-fidelity performance in challenging predictor-free and highly expressive scenarios.

5. Generative AI Use Disclosure

During the preparation of this manuscript, the authors used generative AI tools exclusively for the purpose of language editing and manuscript polishing to improve readability. These tools were not used to generate any core scientific ideas, experimental data, or technical contributions. All authors have thoroughly reviewed and approved the final version of the manuscript, and assume full responsibility for the integrity and entirety of its content.

6. Acknowledgements

This work was supported by National Natural Science Foundation of China (62076144), National Social Science Foundation of China (13&ZD189) and Ant Group.

7. References

- [1] Y. Wang, R. Skerry-Ryan, D. Stanton, Y. Wu, R. J. Weiss, N. Jaitly, Z. Yang, Y. Xiao, Z. Chen, S. Bengio *et al.*, “Tacotron: Towards end-to-end speech synthesis,” in *Proceedings of the Annual Conference of the International Speech Communication Association (INTERSPEECH)*, 2017.
- [2] N. Li, S. Liu, Y. Liu, S. Zhao, and M. Liu, “Neural speech synthesis with transformer network,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2019, pp. 6706–6713.
- [3] V. Popov, I. Vovk, V. Gogoryan, T. Sadekova, and M. Kudinov, “Grad-tts: A diffusion probabilistic model for text-to-speech,” in *Proceedings of the International Conference on Machine Learning (ICML)*. PMLR, 2021, pp. 8599–8608.
- [4] Y. Hou, H. Liu, Y. Wang, Z. Cheng, R. Wu, Q. Gu, Y. Wang, and Y. Wang, “Sova-bench: Benchmarking the speech conversation ability for llm-based voice assistant,” in *Proceedings of the Annual Conference of the International Speech Communication Association (INTERSPEECH)*, 2025.
- [5] J. Francis, J. Gustafsson, and É. Székely, “From static to dynamic: Enhancing aac with generative imagery and zero-shot tts,” in *Proceedings of the Annual Conference of the International Speech Communication Association (INTERSPEECH)*, 2025, pp. 4960–4962.
- [6] W. Zhang, C.-C. Yeh, W. Beckman, T. Raitio, R. Rasipuram, L. Golipour, and D. Winarsky, “Audiobook synthesis with long-form neural text-to-speech,” in *Proceedings of the 12th Speech Synthesis Workshop (SSW) 2023*, 2023.
- [7] W. Ping, K. Peng, A. Gibiansky, S. O. Arik, A. Kannan, S. Narang, J. Raiman, and J. Miller, “Deep voice 3: Scaling text-to-speech with convolutional sequence learning,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2018.
- [8] J. Shen, R. Pang, R. J. Weiss, M. Schuster, N. Jaitly, Z. Yang, Z. Chen, Y. Zhang, Y. Wang, R. Skerry-Ryan *et al.*, “Natural tts synthesis by conditioning wavenet on mel spectrogram predictions,” in *Proceedings of the 2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2018, pp. 4779–4783.
- [9] Y. Song, Z. Chen, X. Wang, Z. Ma, and X. Chen, “Ella-v: Stable neural codec language modeling with alignment-guided sequence reordering,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2025, pp. 25 174–25 182.
- [10] C. Du, Y. Guo, H. Wang, Y. Yang, Z. Niu, S. Wang, H. Zhang, X. Chen, and K. Yu, “Vall-t: Decoder-only generative transducer for robust and decoding-controllable text-to-speech,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2025, pp. 1–5.
- [11] P. Peng, P.-Y. Huang, S.-W. Li, A. Mohamed, and D. Harwath, “Voicecraft: Zero-shot speech editing and text-to-speech in the wild,” in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024, pp. 12 442–12 462.
- [12] X. Tan, J. Chen, H. Liu, J. Cong, C. Zhang, Y. Liu, X. Wang, Y. Leng, Y. Yi, L. He *et al.*, “Naturalspeech: End-to-end text-to-speech synthesis with human-level quality,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 4234–4245, 2024.
- [13] K. Shen, Z. Ju, X. Tan, Y. Liu, Y. Leng, L. He, T. Qin, S. Zhao, and J. Bian, “Naturalspeech 2: Latent diffusion models are natural and zero-shot speech and singing synthesizers,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2023.
- [14] Z. Ju, Y. Wang, K. Shen, X. Tan, D. Xin, D. Yang, Y. Liu, Y. Leng, K. Song, S. Tang *et al.*, “Naturalspeech 3: zero-shot speech synthesis with factorized codec and diffusion models,” in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.
- [15] Y. Ren, Y. Ruan, X. Tan, T. Qin, S. Zhao, Z. Zhao, and T.-Y. Liu, “Fastspeech: Fast, robust and controllable text to speech,” *Advances in Neural Information Processing Systems*, 2019.
- [16] K. Peng, W. Ping, Z. Song, and K. Zhao, “Parallel neural text-to-speech,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2020.
- [17] J. Kim, S. Kim, J. Kong, and S. Yoon, “Glow-tts: A generative flow for text-to-speech via monotonic alignment search,” in *Proceedings of the Advances in Neural Information Processing Systems (NeurIPS)*, 2020, pp. 8067–8077.
- [18] J. Kim, J. Kong, and J. Son, “Conditional variational autoencoder with adversarial learning for end-to-end text-to-speech,” in *International Conference on Machine Learning (ICML)*. PMLR, 2021, pp. 5530–5540.
- [19] M. McAuliffe, M. Socolof, S. Mihuc, M. Wagner, and M. Sonderegger, “Montreal forced aligner: Trainable text-speech alignment using kaldi,” in *Proceedings of the Annual Conference of the International Speech Communication Association (INTERSPEECH)*, 2017, pp. 498–502.
- [20] S. E. Eskimez, X. Wang, M. Thakker, C. Li, C.-H. Tsai, Z. Xiao, H. Yang, Z. Zhu, M. Tang, X. Tan *et al.*, “E2 tts: Embarrassingly easy fully non-autoregressive zero-shot tts,” in *Proceedings of the 2024 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2024, pp. 682–689.
- [21] Y. Chen, Z. Niu, Z. Ma, K. Deng, C. Wang, J. JianZhao, K. Yu, and X. Chen, “F5-tts: A fairytale that fakes fluent and faithful speech with flow matching,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025, pp. 6255–6271.
- [22] Y. Wang, H. Zhan, L. Liu, R. Zeng, H. Guo, J. Zheng, Q. Zhang, X. Zhang, S. Zhang, and Z. Wu, “Maskgct: Zero-shot text-to-speech with masked generative codec transformer,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.
- [23] I. Elias, H. Zen, J. Shen, Y. Zhang, Y. Jia, R. Skerry-Ryan, and Y. Wu, “Parallel tacotron 2: A non-autoregressive neural tts model with differentiable duration modeling,” in *Proceedings of the Annual Conference of the International Speech Communication Association (INTERSPEECH)*, 2021.
- [24] J. Li, X. Dong, Y. Zang, Y. Cao, J. Wang, and D. Lin, “Beyond fixed: Variable-length denoising for diffusion large language models,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026.
- [25] Z. Wu, L. Zheng, Z. Xie, J. Ye, J. Gao, Y. Feng, Z. Li, V. W., G. Zhou, and L. Kong, “Dreamon: Diffusion language models for code infilling beyond fixed-size canvas,” 2025. [Online]. Available: <https://hkunlp.github.io/blog/2025/dreamon>
- [26] S. Sahoo, M. Arriola, Y. Schiff, A. Gokaslan, E. Marroquin, J. Chiu, A. Rush, and V. Kuleshov, “Simple and effective masked diffusion language models,” in *Proceedings of the Advances in Neural Information Processing Systems (NeurIPS)*, 2024, pp. 130 136–130 184.